

Title:Student Record Management System Using Singly Linked List

1. Problem Statement

To develop a Student Record Management System using a Singly Linked List in C++. The system stores student information such as Roll Number, Name, Branch, and Marks. It allows the user to add and display student records dynamically using linked list operations.

2. Objectives

- To understand the concept of Singly Linked List.
- To implement dynamic memory allocation in C++.
- To add and store student records dynamically.
- To display student records efficiently.
- To understand the practical application of linked lists in data management.
- To improve programming and problem-solving skills.

3. Algorithm

Step 1: Start the program.

Step 2: Define a structure Student containing:

Roll Number

Name

Branch

Marks

Pointer next

Step 3: Initialize head = NULL.

Step 4: Create a new student node dynamically.

Step 5: Accept Roll Number, Name, Branch and Marks from the user.

Step 6: If the linked list is empty, make the new node the head.

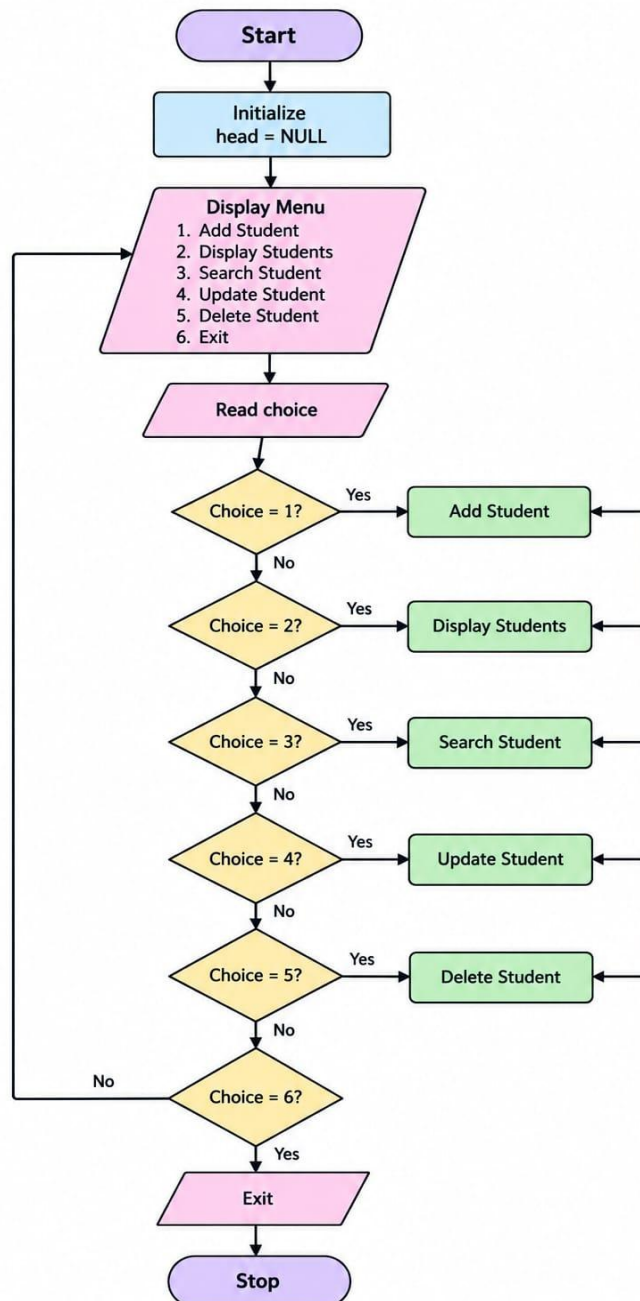
Step 7: Otherwise, traverse the list and add the new node at the end.

Step 8: To display records, start from head and traverse the linked list.

Step 9: Display the details of each student.

Step 10: Stop the program.

4. Flowchart



5. Conclusion

The Student Record Management System using Singly Linked List successfully demonstrates the practical implementation of linked list concepts in C++. It provides a simple and dynamic method for storing and displaying student records. This project helped in understanding nodes, pointers, dynamic memory allocation, traversal, and linked list operations. It also improved our practical programming and problem-solving skills.